

Answers to the mid-term examination on XML

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1 Queues

A *queue* is a stack where the items are popped at the **bottom**, not at the top. Pushing an item in a queue is called *enqueueing*, whereas the special pop is called *dequeueing*.

ENQUEUE $\rightarrow$

	a	b	c	d	e	
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 $\rightarrow$ DEQUEUE

PUSH, POP $\leftrightarrow$

	a	b	c	d	e	
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We want to implement queues with two stacks:

ENQUEUE $\rightarrow$

	a	b	c	
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d	e	
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 $\rightarrow$ DEQUEUE

So ENQUEUE is PUSH on the first stack and DEQUEUE is POP on the second. If the second stack is empty, we swap the stacks and reverse the (new) second:

ENQUEUE $\rightarrow$

	a	b	c	
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 $\rightarrow$ DEQUEUE ???

ENQUEUE $\rightarrow$

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a	b	c	
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 $\rightarrow$ DEQUEUE

Data. Let $\text{QUEUE}(S, T)$ be the queue made of the stacks S and T ; let $\text{PUSH}(e, S)$ be the stack S with item e on top; let EMPTY be the empty stack.

Functions. Let $\text{ENQUEUE}(e, Q)$ rewrite to the queue Q where item e is added; let $\text{DEQUEUE}(Q)$ rewrite to the pair (e, Q') where e is the item on the head of queue Q and Q' is the remaining queue.

Question. Define $\text{ENQUEUE}(e, Q)$ and $\text{DEQUEUE}(Q)$, where Q is a queue.

Answer.

$$\begin{aligned}
\text{DEQUEUE}(\text{QUEUE}(S, \text{PUSH}(e, T))) &\xrightarrow{1} (e, \text{QUEUE}(S, T)) \\
\text{DEQUEUE}(\text{QUEUE}(S, \text{EMPTY})) &\xrightarrow{2} \text{DEQUEUE}(\text{QUEUE}(\text{EMPTY}, \text{REV}(S))) \\
\text{ENQUEUE}(e, \text{QUEUE}(S, T)) &\xrightarrow{3} \text{QUEUE}(\text{PUSH}(e, S), T)
\end{aligned}$$

But there is a problem with DEQUEUE if the queue is empty. Indeed, in this case we have

$$\begin{aligned}
&\text{DEQUEUE}(\text{QUEUE}(\text{EMPTY}, \text{EMPTY})) \\
&\quad \xrightarrow{2} \text{DEQUEUE}(\text{QUEUE}(\text{EMPTY}, \text{REV}(\text{EMPTY}))) \\
&\quad \rightarrow \text{DEQUEUE}(\text{QUEUE}(\text{EMPTY}, \text{EMPTY}))
\end{aligned}$$

which is a never-ending loop! The fix consists in forbidding the use of rule 2 when the queue is empty. One easy way is simply to say

$$\begin{aligned}
&\text{DEQUEUE}(\text{QUEUE}(S, \text{EMPTY})) \xrightarrow{2} \text{DEQUEUE}(\text{QUEUE}(\text{EMPTY}, \text{REV}(S))) \\
&\quad \text{if } S \neq \text{EMPTY}
\end{aligned}$$

Another way is to add an error-case rule like

$$\text{DEQUEUE}(\text{QUEUE}(\text{EMPTY}, \text{EMPTY})) \xrightarrow{4} \text{ERROR}$$

and prefer it over rule 3.

2 Stacks

Question. Define $\text{BOTTOM}(S)$ be the pair (e, T) where e is the item at the bottom of stack S and T is S without e .

Answer.

$$\begin{aligned}
&\text{BOTTOM}(S) \rightarrow \text{BOTTOM}'(\text{REV}(S)) \\
&\text{BOTTOM}'(\text{PUSH}(e, T)) \rightarrow (e, \text{REV}(T))
\end{aligned}$$

Note. The two questions are related. Indeed, BOTTOM provides a means to use a stack as a queue: ENQUEUE is simply PUSH and DEQUEUE is BOTTOM ! But BOTTOM is expensive: in order to get the element at the bottom, i.e. dequeue, we have to reverse the whole stack two times (see S and T), except one element. But in the first question, DEQUEUE is cheap: if the second stack is not empty, it is simply equivalent to a POP (one rewrite). If it is empty, then the cost is to reverse the first stack, i.e. to traverse *only once* all the elements — instead of two times, every time, with BOTTOM . There is a direct way to make queues (that is, without using stacks). We need here an empty queue, noted Q_0 , but **no** QUEUE .

Example. The queue containing only item e is $\text{ENQUEUE}(e, Q_0)$. The queue containing item e and then f is $\text{ENQUEUE}(f, \text{ENQUEUE}(e, Q_0))$. We have

$$\text{DEQUEUE}(\text{ENQUEUE}(f, \text{ENQUEUE}(e, Q_0))) \rightarrow \cdots \rightarrow (\text{ENQUEUE}(f, Q_0), e)$$

Question. Give the rewrite rules defining DEQUEUE .

Answer.

$$\begin{aligned} \text{DEQUEUE}(\text{ENQUEUE}(e, Q_0)) &\rightarrow (Q_0, e) \\ \text{DEQUEUE}(\text{ENQUEUE}(e, Q)) &\rightarrow (\text{ENQUEUE}(e, Q'), f) \\ &\text{where } q \neq Q_0 \text{ and } \text{DEQUEUE}(Q) \rightarrow (Q', f) \end{aligned}$$